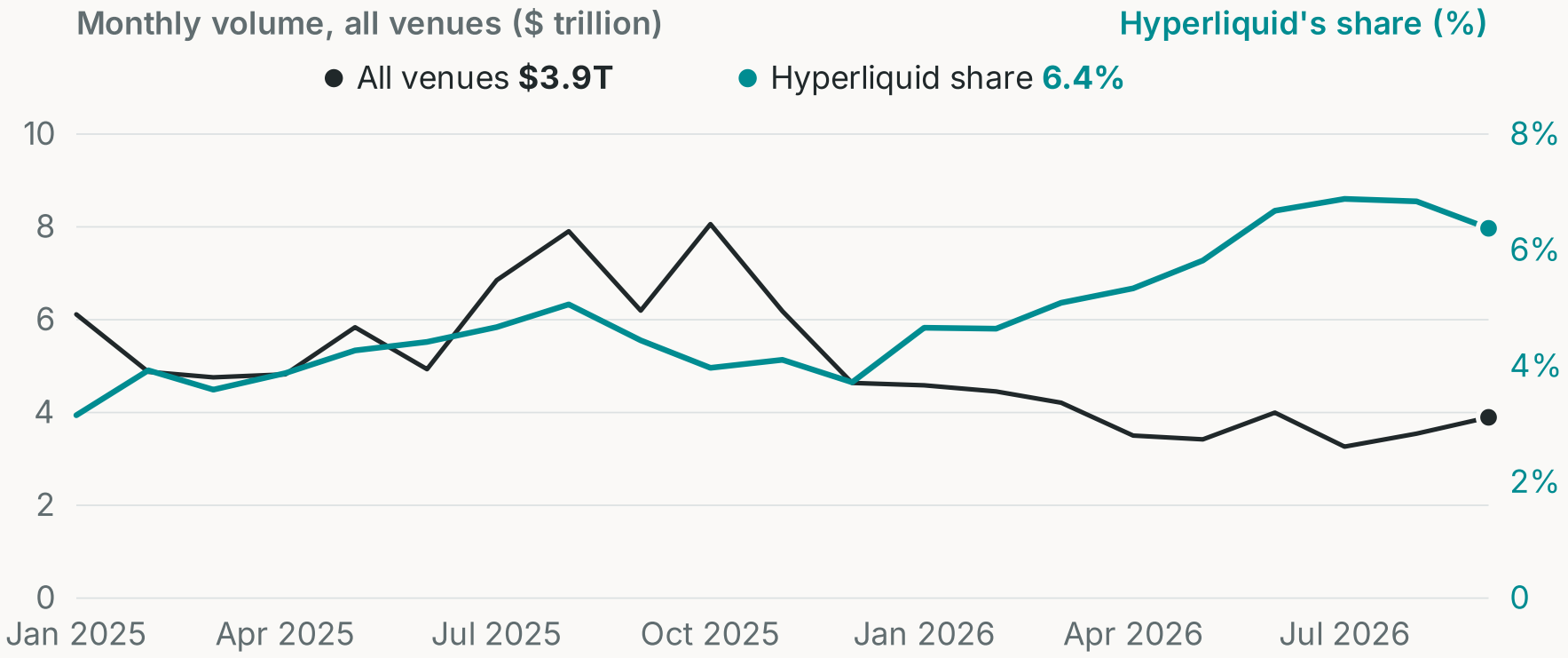
Who Loses and Why?

Market Microstructure of Perpetual Futures

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Shouqiao Wang · Ciamac Moallemi *Columbia University*

Perpetual Futures Trade Trillions a Month

- **A perpetual future** is a futures contract with no expiry: traders go long or short, with leverage.
- **Coming onshore:** the first U.S. bitcoin perpetual was approved in 2026.



Hyperliquid Shows the Trader, Not Just the Trade

Top 10 coins · 2025.03–2026.06 · $2.95 trillion volume · 729,540 wallets

- **Every fill names the wallet**, its order type, and whether it opens, closes or is a liquidation.

TIME (UTC)	WALLET	COIN	SIDE	PRICE	SIZE	ORDER TYPE	ROLE	DIRECTION	LIQUIDATION
14:03:07.212	0x7c2…e41	SOL	Buy	180.03	$35,000	FrontendMarket	Taker	Open long	No
14:03:07.212	0x91a…0bd	SOL	Sell	180.03	$35,000	Alo	Maker	Open short	No
14:03:07.950	0x4de…772	ETH	Sell	4,102.5	$12,400	Ioc	Taker	Close long	No
14:03:08.105	0x2b8…c19	BTC	Sell	108,214	$61,800	Liquidation	Taker	Close long	Yes

Five Questions, Five Answers

QUESTION	ANSWER
Who trades?	Market makers, front-end retail traders, algorithmic takers
Who loses?	**Front-end retail lose $1.11B; market makers gain $211M**
Which retail?	Small accounts at high leverage, who deposit again as they lose
Why do they lose?	Fees $725M > liquidation $226M > trading loss $154M
What helps?	Fee schedule and leverage limits

What a Retail Trader Sees

Four order types; the front-end market order is the retail signal

HYPERLIQUID

Replay · real 5-minute BTC candles and order-book snapshots · book from Hyperliquid data

↻ REPLAY

BTC-USDC 20x

Mark 110,592 | 24h Change -1.90% | 24h Volume $3,142,527,612 | 5m · Oct 11, 23:55 UTC



Order Book BTC

Price	Size (BTC)	Total (BTC)
110,600	0.1355	1.4839
110,599	0.9033	1.3484
110,597	0.0002	0.4451
110,595	0.0972	0.4449
110,593	0.3477	0.3477
Spread	1	0.001%
110,592	7.6081	7.6081
110,591	0.9033	8.5114
110,590	0.0648	8.5761
110,589	0.0272	8.6034
110,588	1.8426	10.4460

Cross | 20x

Market | Limit | Pro

Buy / Long | Sell / Short

Available to Trade 1,000.00 USDC

Size 0.05 BTC

Size only: it fills at the asks, now. Sent from this page, the order is stamped FrontendMarket.

Buy / Long

Order Value $5,530

Fees 0.0450% / 0.0150%

Maker Orders Wait on the Book; a Taker Crosses the Spread

ORDER BOOK · HYPE

PRICE	SIZE ($)	TOTAL ($)
100.03	$6,000	$30,000
100.02	$8,000	$24,000
100.01	$16,000	$16,000
spread		
99.99	$13,000	$13,000
99.98	$10,000	$23,000
99.97	$9,000	$32,000

Alo buy at 99.99

Accepted: it waits on the book as a maker.

Alo never crosses the spread.

Three Trader Groups, Told Apart by How They Trade

Market makers

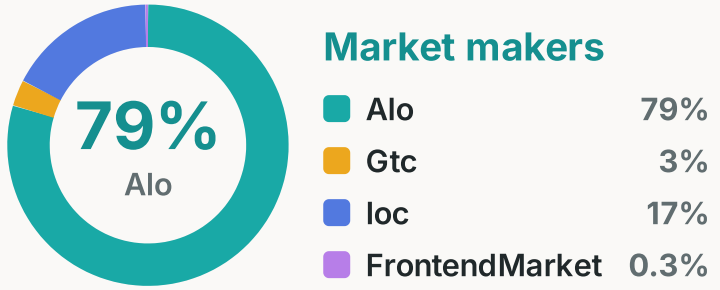
Quote both sides all day with Alo orders.

Front-end retail

Trade by hand on the website or app.

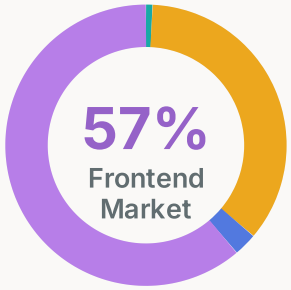
Algorithmic takers

Trade at machine speed with loc and Gtc.



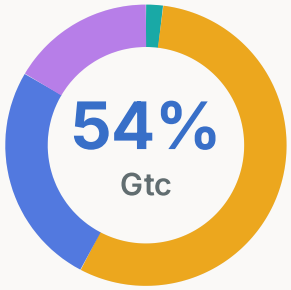
Market makers

Alo	79%
Gtc	3%
loc	17%
FrontendMarket	0.3%



Front-end retail

Alo	0.7%
Gtc	33%
loc	3%
FrontendMarket	57%



Algorithmic takers

Alo	2%
Gtc	54%
loc	25%
FrontendMarket	16%

Order Types Tell the Three Groups Apart

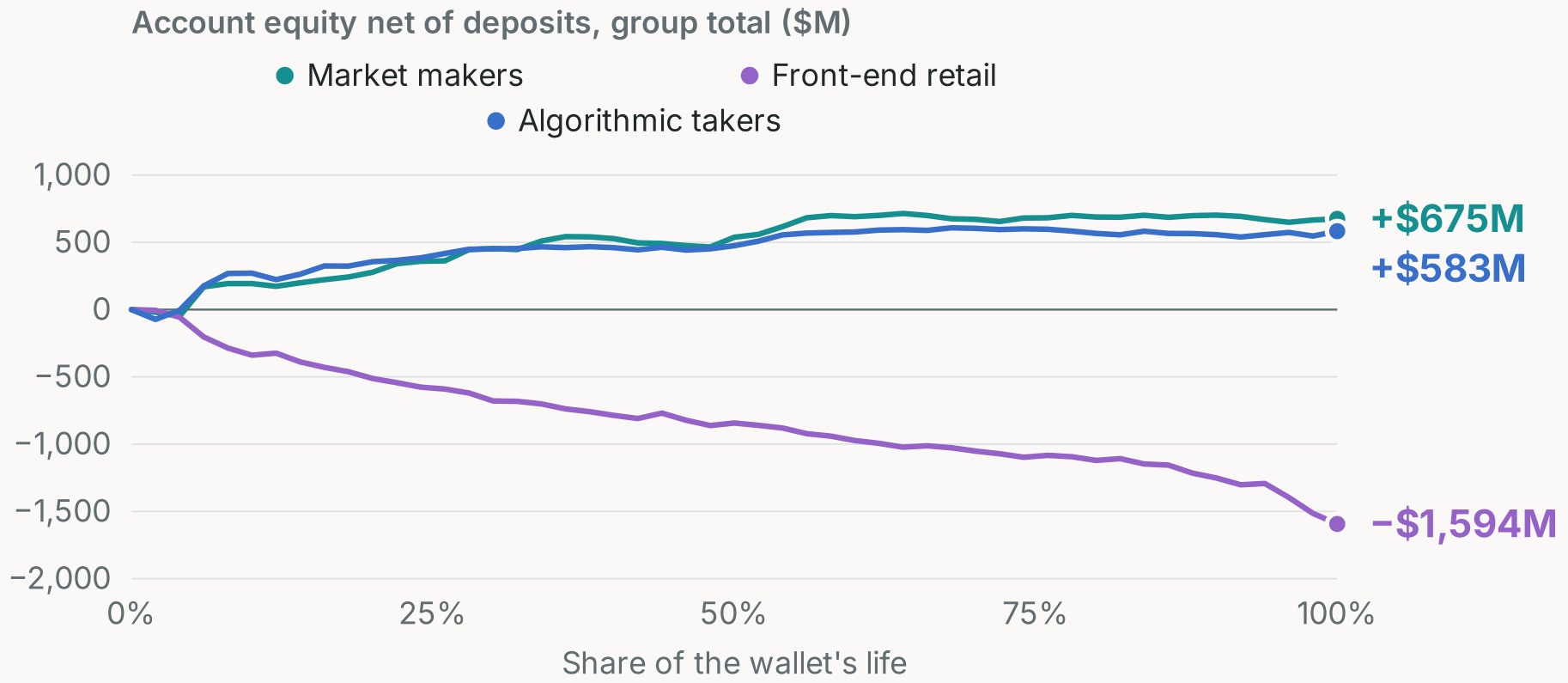
ORDER TYPE	ROLE	WHAT IT DOES
Alo	maker	Rests on the book; rejected if it would cross the spread
Gtc	maker or taker	Trades at once up to its limit price; the rest waits on the book
Ioc	taker	Trades at once up to its limit price; the rest is cancelled
FrontendMarket	taker	A market order sent from Hyperliquid's website or app: trades at once until the size is filled

Three Groups Trade Very Differently

- **Retail start small, use high leverage** and trade mostly as takers with front-end market orders.

GROUP	TAKER RATIO	OPENING LEVERAGE, MEDIAN	STARTING EQUITY · P50 / P95
Market makers	20%	1.4×	$26,052 / $1.64M
Front-end retail	**85%**	**9.0×**	**$306 / $50,342**
Algorithmic takers	69%	3.7×	$386 / $53,485

Retail Accounts Lose $1.6 Billion Net of Deposits



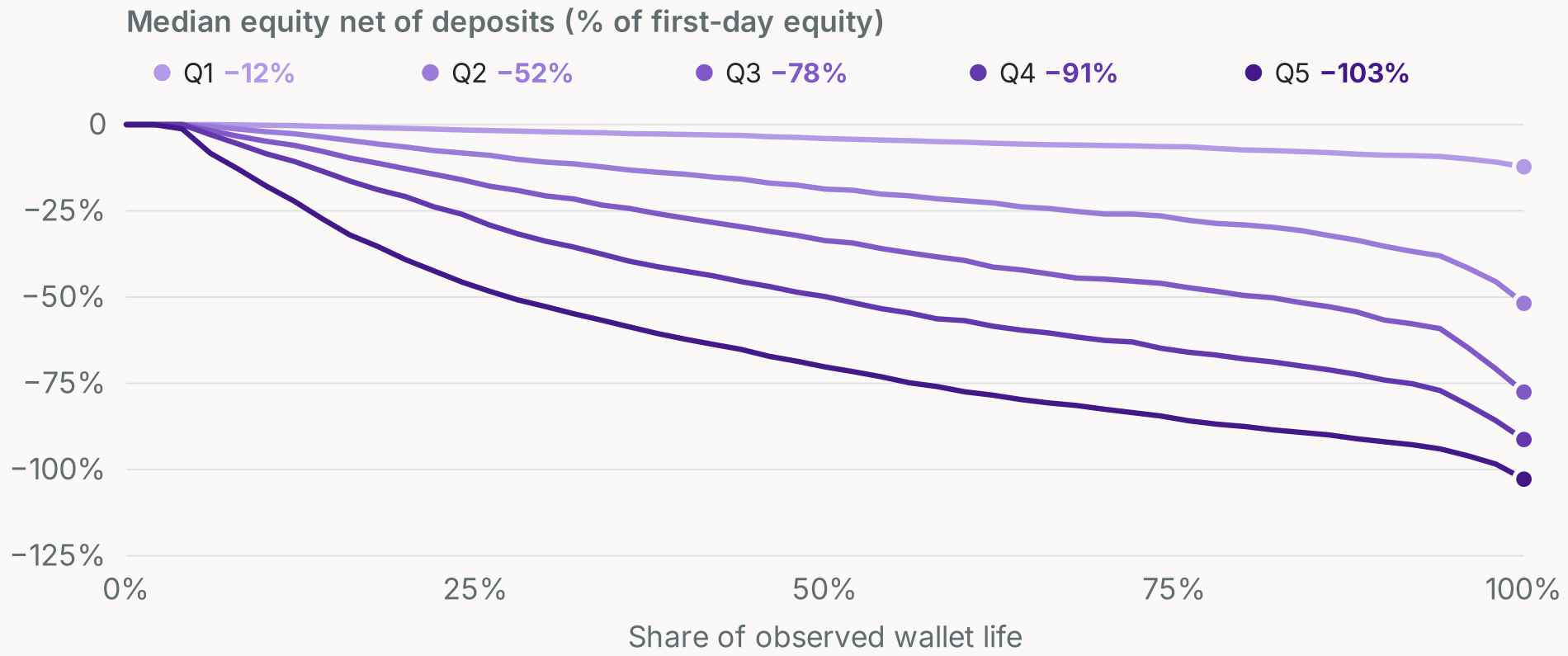
- **The typical retail wallet** loses two-thirds of what it starts with.

Which Retail Loses

- **High-leverage retail start with less equity** and lose it all.

RETAIL COHORT, MEDIAN WALLET	OPENING LEVERAGE	STARTING EQUITY	AVERAGE TRADE SIZE	CASH FLOW AT END OF LIFE
Q1 (lowest)	2.6×	$186	$250	−12%
Q2	5.5×	$195	$899	−52%
Q3	10.0×	$177	$1,105	−78%
Q4	15.8×	$99	$1,175	−91%
Q5 (highest)	**25.1×**	**$33**	$1,095	**−103%**

Gambler's Ruin: They Lose Their Stake and Deposit Again



- **The typical Q5 wallet** loses all it starts with, deposits again, and loses more.
- **Half of Q5 wallets** lose at least their stake, against one in six in Q1.

We Split Retail Loss into Three Channels

1. **Fees**
   what retail pay the venue

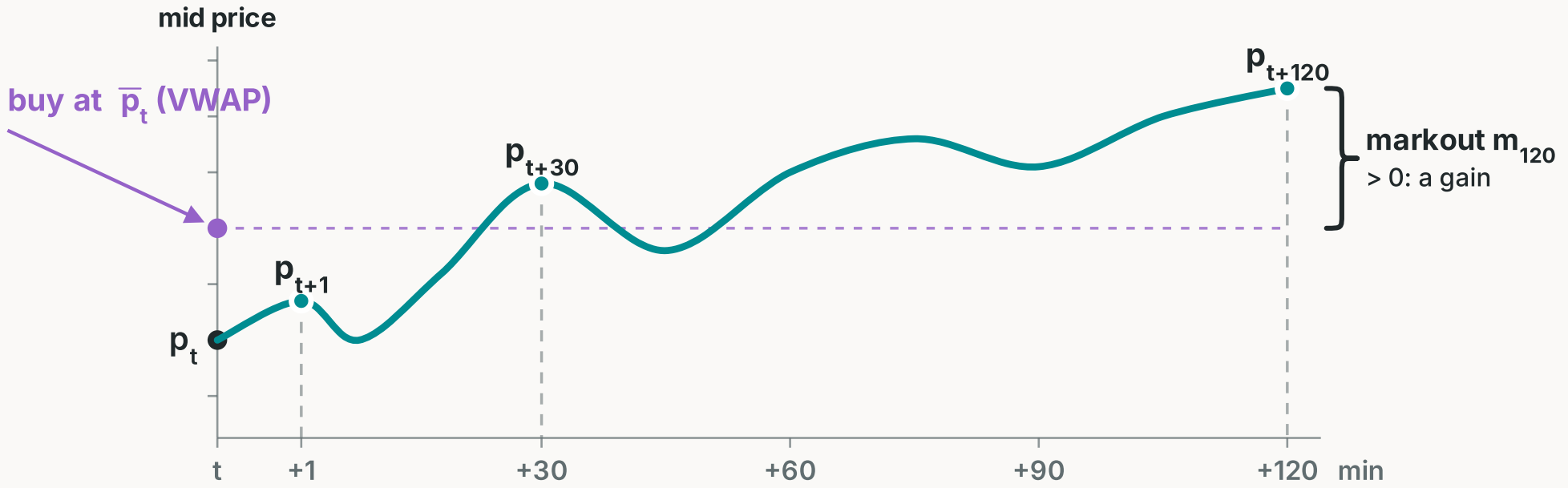
2. **Liquidation**
   positions the platform closes when equity runs out

3. **Trading loss**
   execution cost and timing of voluntary trades
   Measured by markout: how the price moves after the trade.

Markout: How the Price Moves After a Trade

$$m_h = s \times \frac{p_{t+h} - \bar{p}_t}{\bar{p}_t} \times 10^4$$

m_h = markout, bps s = +1 buy, −1 sell $\bar{p}_t$ = trade price (VWAP) at time t p_{t+h} = mid price h minutes later



Retail Lose $1.11 Billion; the Venue Collects $830 Million in Fees

- **Market makers gain,** algorithmic takers break even, retail lose.
- **Fees are the largest channel** of the retail loss.

GROUP	TRADING LOSS	LIQUIDATION	FUNDING	FEES	TOTAL
Market makers	+$296M	−$38M	+$4M	−$50M	+$211M
Front-end retail	−$154M	−$226M	−$5M	**−$725M**	**−$1.11B**
Algorithmic takers	+$40M	+$11M	+$1M	−$55M	−$4M

Channel 1: Fees

1. **Fees**
   what retail pay the venue
2. Liquidation
   positions the platform closes when equity runs out
3. Trading loss
   execution cost and timing of voluntary trades

How the Fee Is Set: Staking Tier, Then Volume, Then Role

1 HYPE staked

The staking tier sets the discount on every rate.

2 Trading volume

Within the tier, more volume means a lower fee rate.

3 Taker or maker

Takers pay the higher rate. The largest makers earn a rebate.

VOLUME TIER	14-DAY WEIGHTED VOLUME	TAKER BASE RATE	MAKER BASE RATE
Tier 0	$0–$5M	4.5 bps (0.045%)	1.5 bps (0.015%)

Fees: Retail Pay 87% of Fees on 42% of Volume

- **Retail trade mostly as takers,** and takers pay the higher rate.
- **Retail sit in the highest fee tier:** little HYPE staked, low volume.

	TAKER FEE (BPS)	MAKER FEE (BPS)	TAKER RATIO	VOLUME FROM NON-STAKERS
Market makers	1.71	**rebate 0.22**	20%	24%
Front-end retail	**3.32**	0.80	**85%**	**74%**
Algorithmic takers	2.87	0.55	69%	43%

Channel 2: Liquidation

1. Fees
   what retail pay the venue
2. **Liquidation**
   positions the platform closes when equity runs out
3. Trading loss
   execution cost and timing of voluntary trades

Liquidation: Retail Bear It, and It Is the Costliest Exit

- **Retail bear almost all liquidations:** 97.6% of liquidation volume outside October 10.
- **Liquidations are large, pay the widest half-spread and have the worst timing:** they land near the bottom of a fall, and the price reverts soon after.

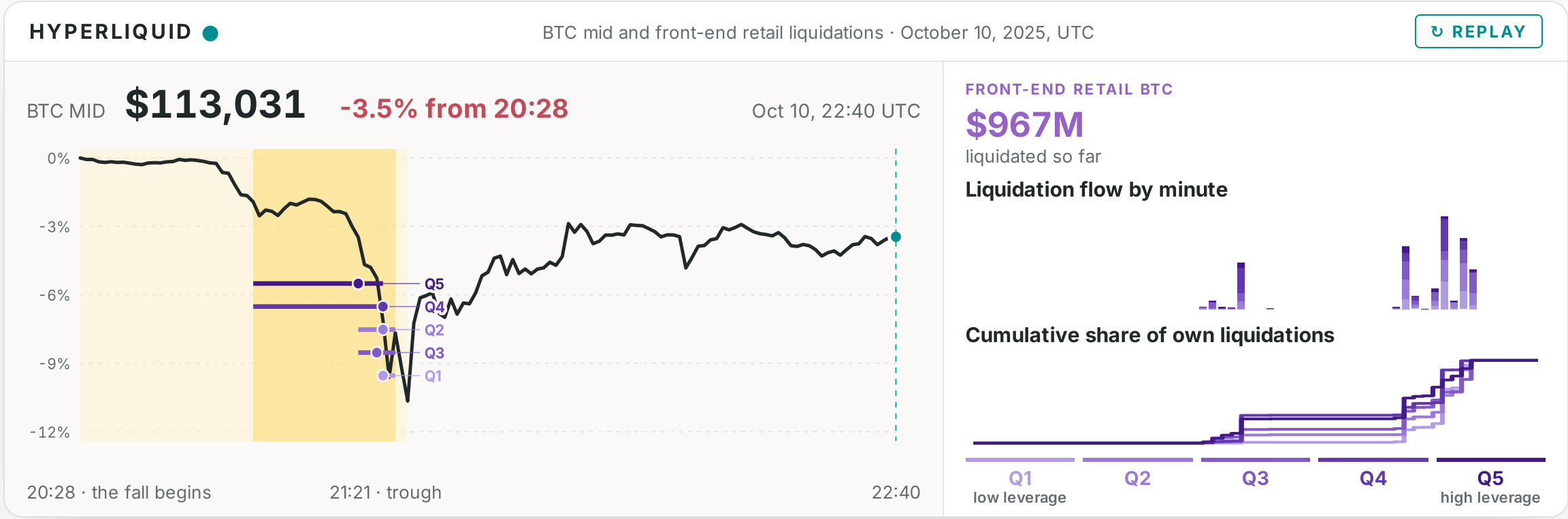
THREE WAYS TO CLOSE A TRADE	TRADE SIZE	HALF-SPREAD (BPS)	TIMING: PRICE IMPACT AT 2 H (BPS)
Gain-realizing close	$7.0k	0.69	+2.54
Loss-realizing close	$7.7k	0.92	+0.66
Liquidation	**$31.1k**	**15.16**	**−28.48**

Leverage Sets How Often; Cascades Set How Much

- **Q5 is liquidated often,** after small moves.
- **Q1 is liquidated rarely,** but most of its liquidation loss comes on October 10.

RETAIL COHORT	LIQUIDATED SHARE OF CLOSES	WALLETS EVER LIQUIDATED	POSITION AGE, MEDIAN	PRICE MOVE BEFORE	SHARE ON OCT 10	LOSS
Q1	0.85%	6%	**5.6 days**	−13.6%	**44%**	**$67M**
Q2	2.24%	17%	64 h	−8.6%	41%	$77M
Q3	2.50%	24%	28 h	−5.0%	29%	$44M
Q4	4.22%	32%	14 h	−2.8%	14%	$33M
Q5	**8.78%**	**39%**	6.6 h	−1.9%	5.5%	$7M

Low Leverage Is Liquidated Last, at the Worst Price

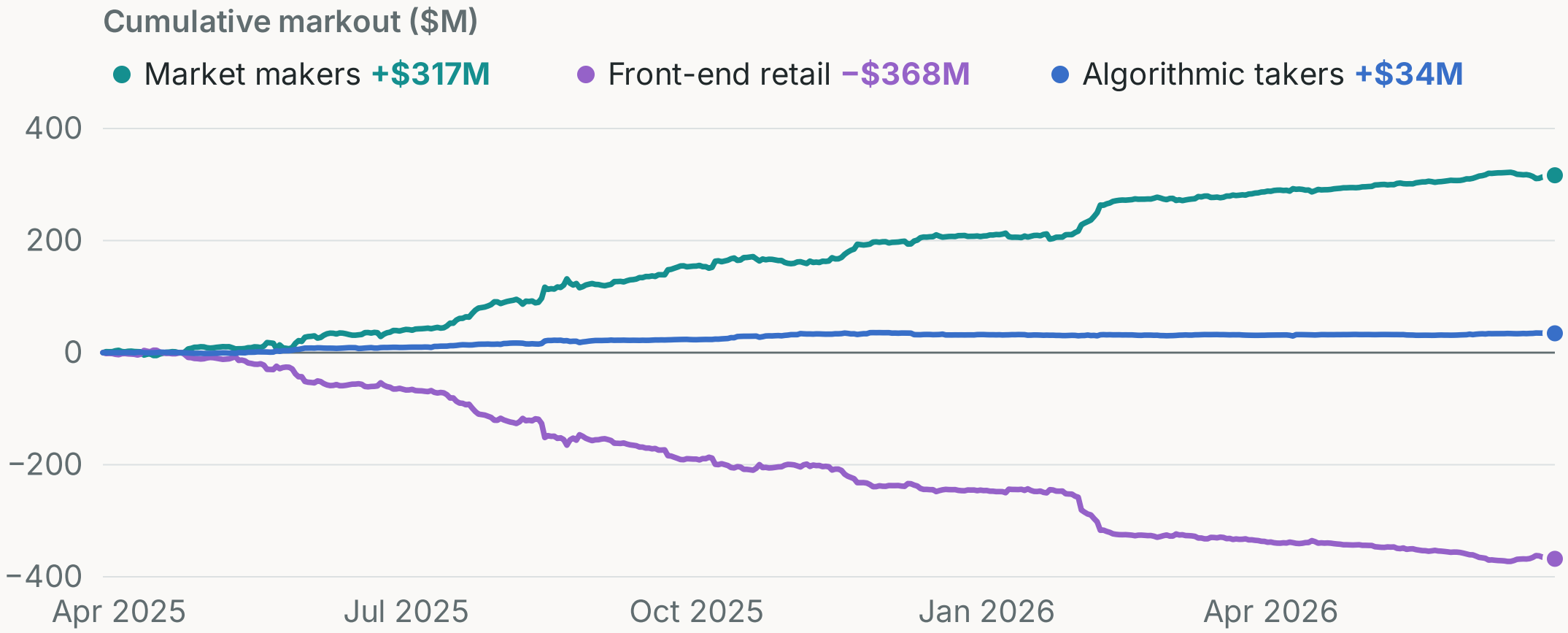


- **High leverage goes first,** after small moves.
- **Low leverage goes near the trough;** BTC then recovers, so they lose most.

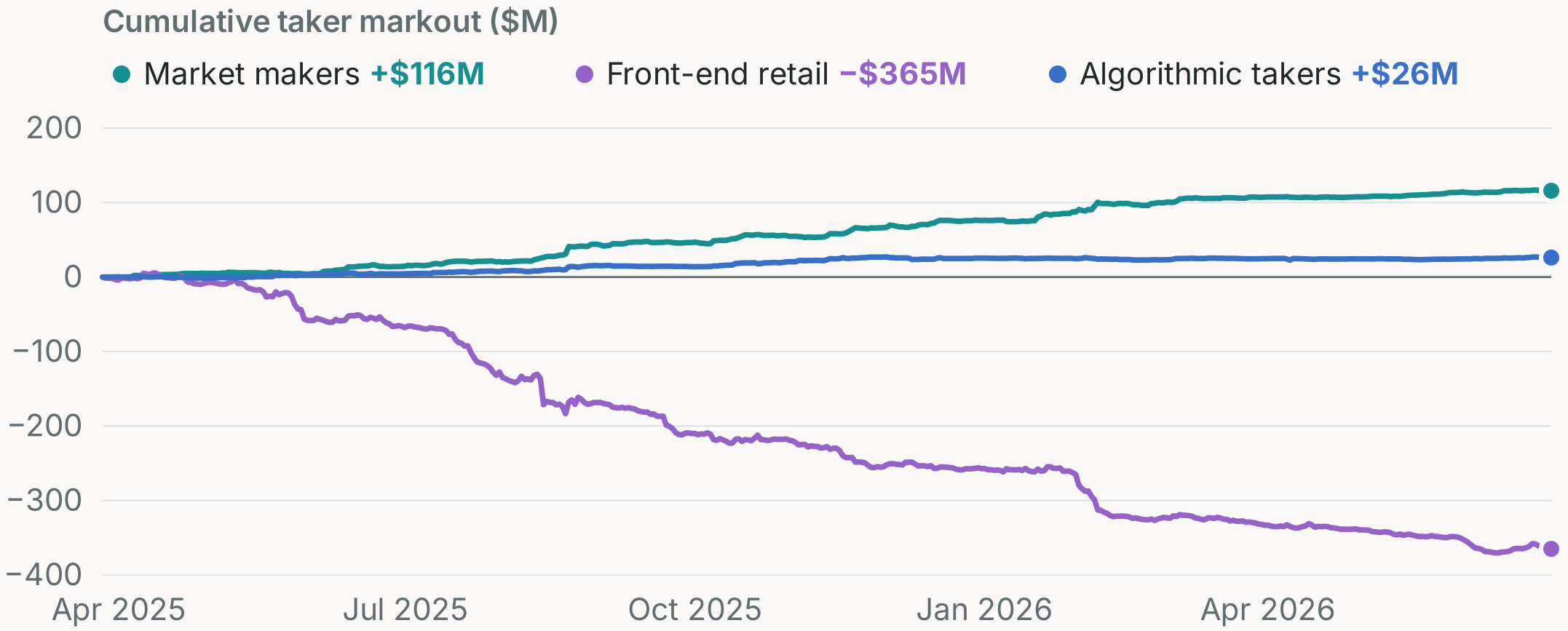
Channel 3: Trading Loss

1. **Fees**
   what retail pay the venue
2. **Liquidation**
   positions the platform closes when equity runs out
3. **Trading loss**
   execution cost and timing of voluntary trades

Retail's Loss Accumulates Over Time

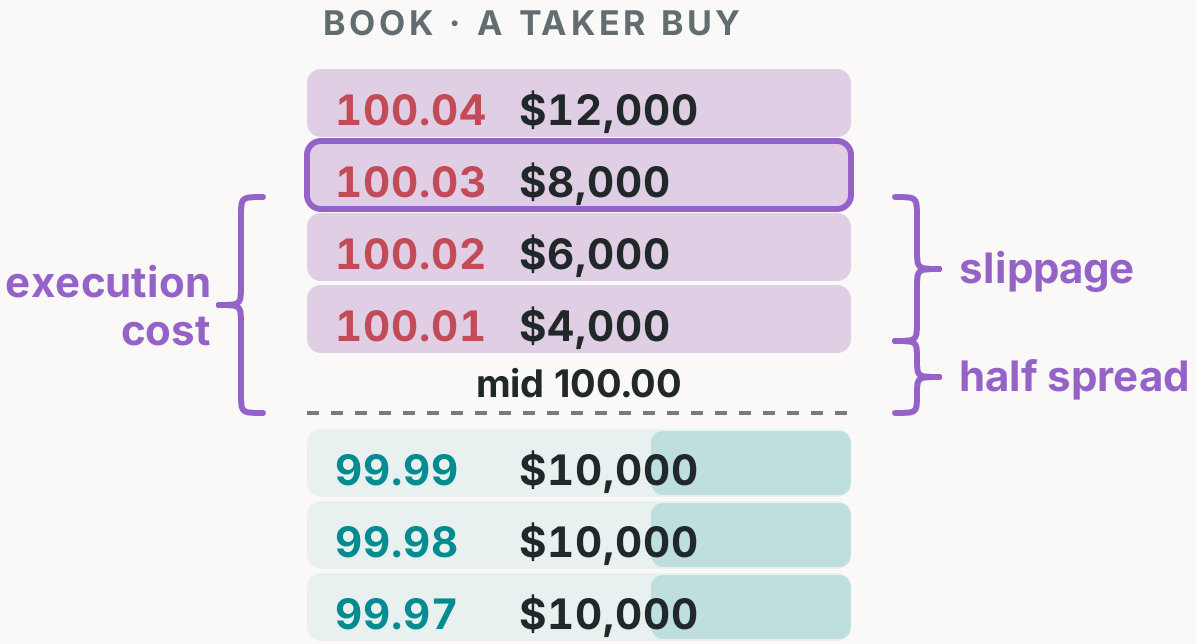


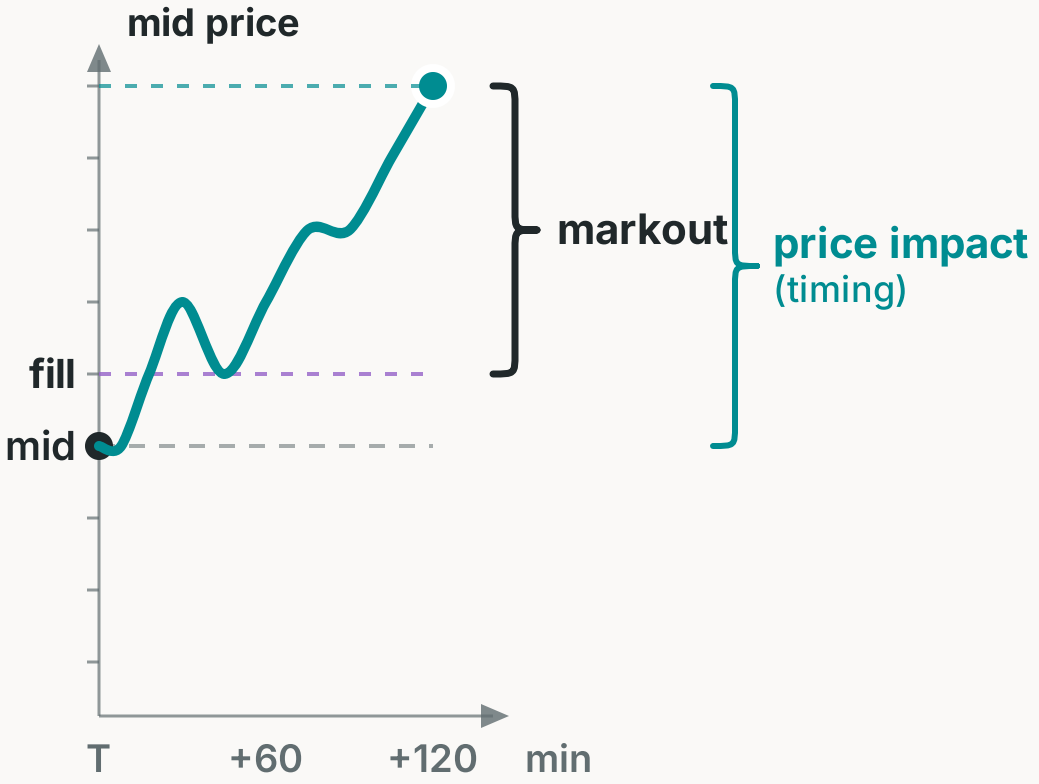
The Loss Is Concentrated on the Taker Side



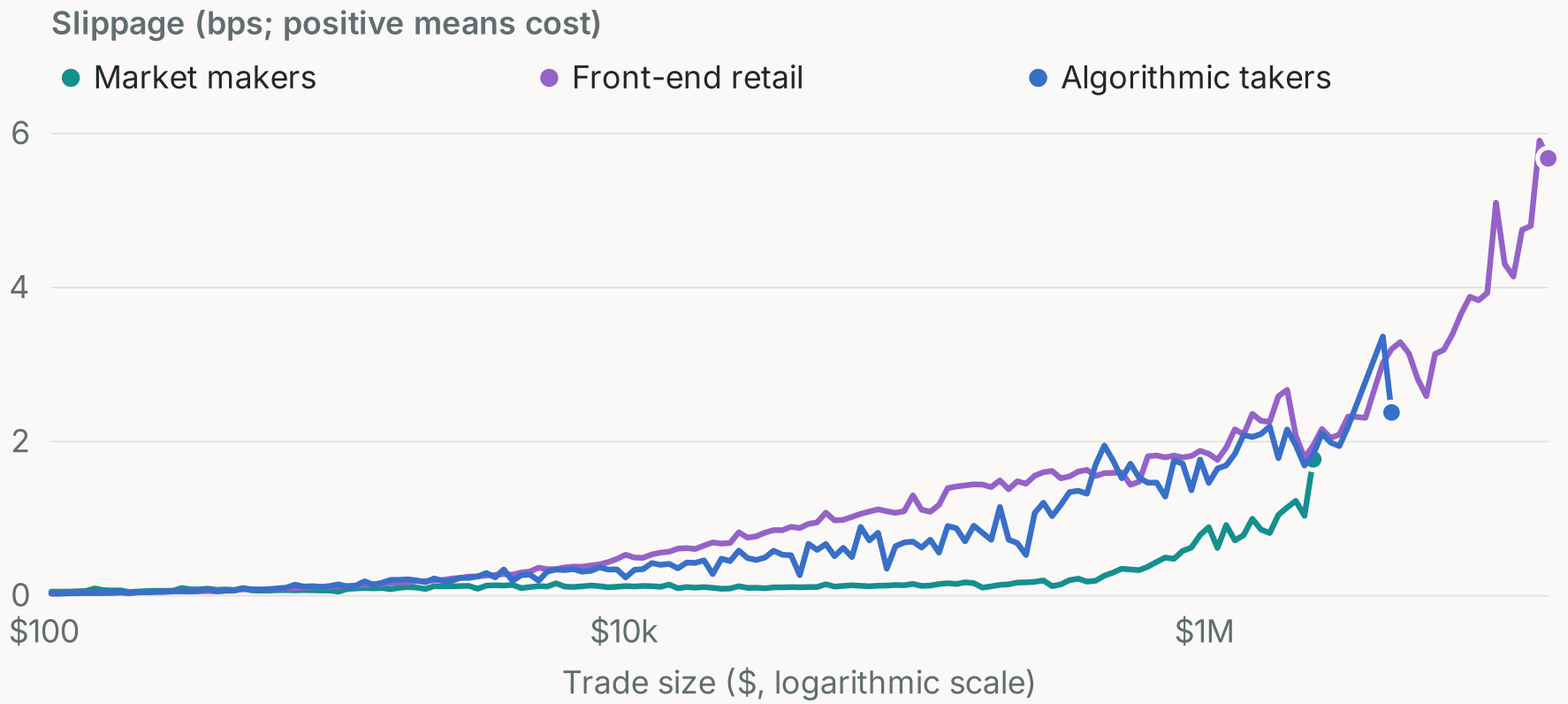
Taker Markout = Execution Cost + Timing

Markout = Price impact + Execution cost | Execution cost = Half spread + Slippage





Execution Cost: Retail Walk the Book



- **Same half-spread for all;** retail pay in slippage.
- **Market makers control slippage** with loc limit orders; retail do not.

The Same Order Walks Further Through a Thin Book

THE SAME BUY OF $30,000 , AT TWO DIFFERENT MOMENTS

Front-end retail

a thin book

FrontendMarket order: size only

ASKS	SIZE
100.10	$16,000
100.05	$8,000
100.03	$6,000

Walks 3 levels: more slippage

Market maker

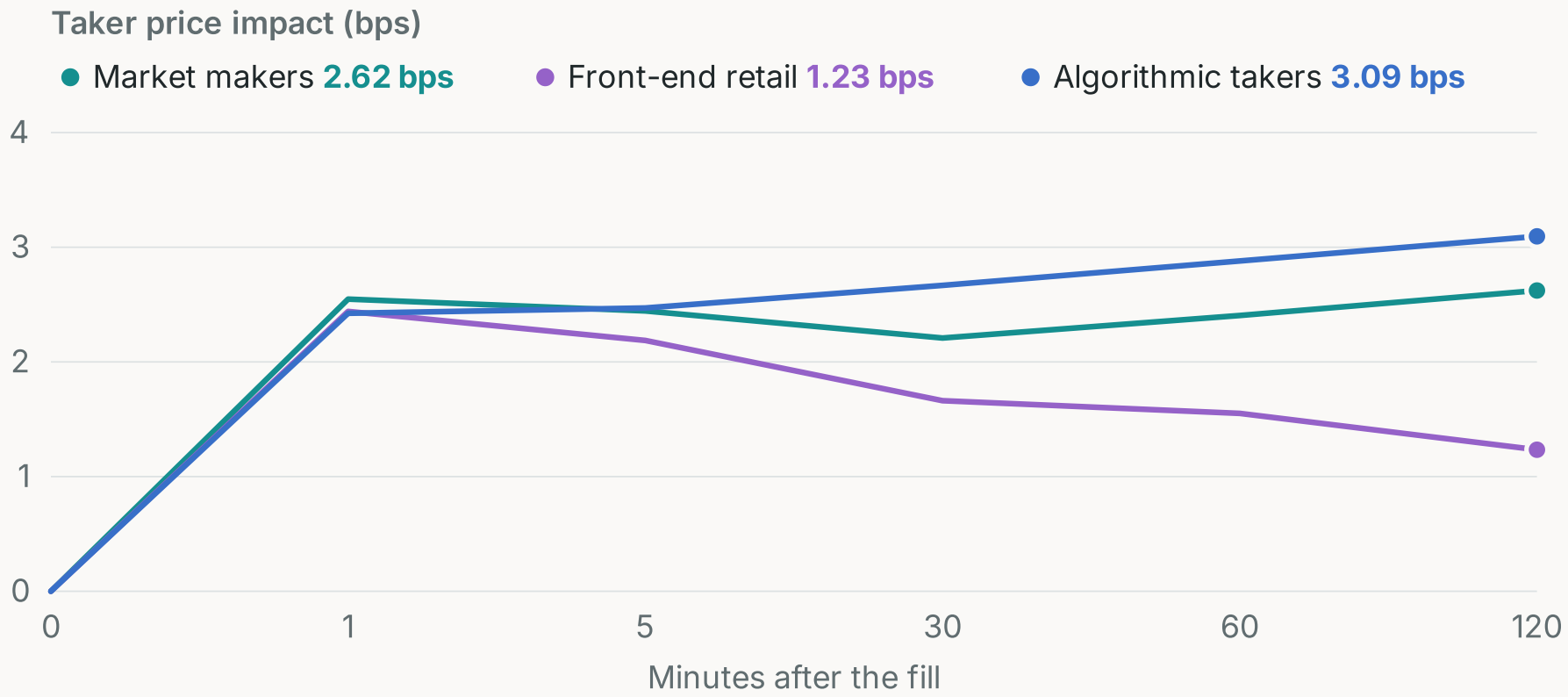
a thick book

loc limit order: size and a price cap

ASKS	SIZE
100.03	$12,000
100.02	$15,000
100.01	$15,000

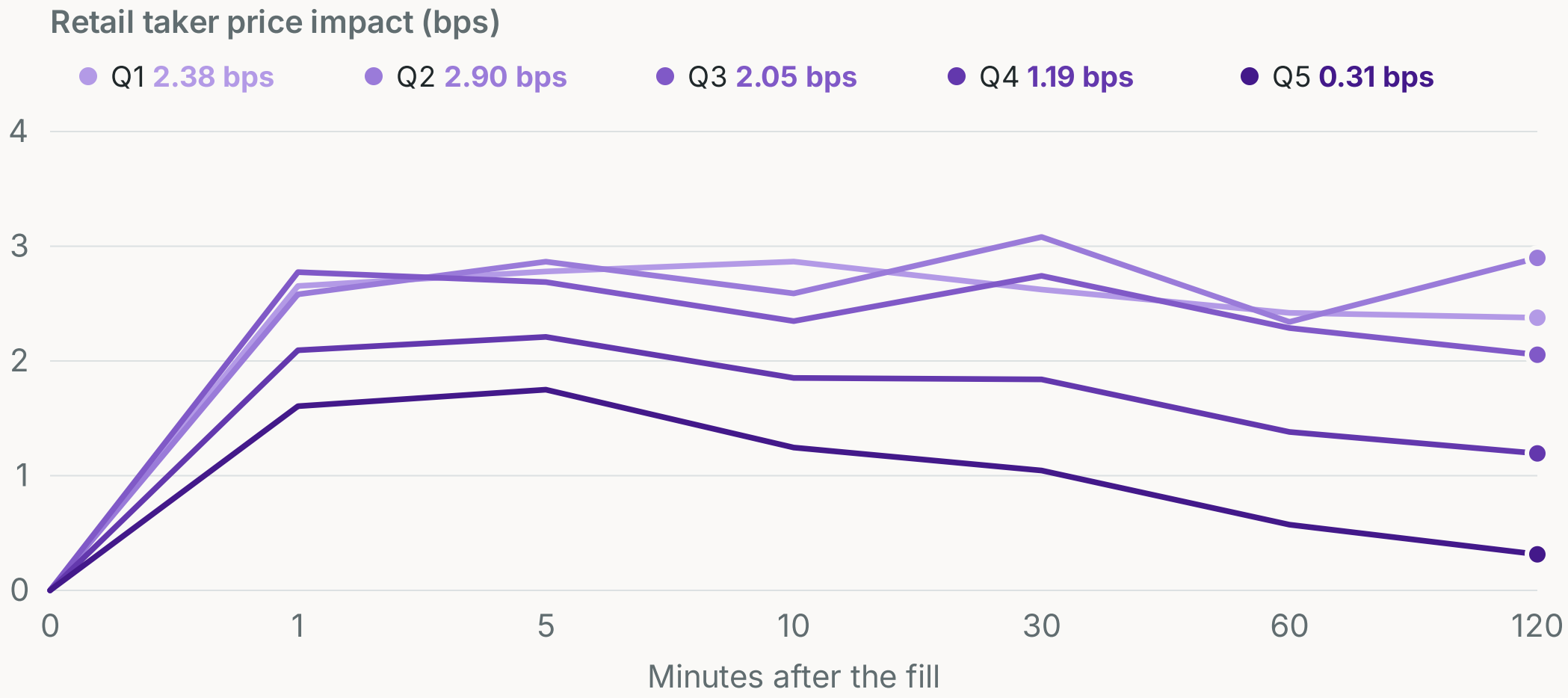
Fills in 2 levels: less slippage

Timing: Retail's Price Impact Halves Within Two Hours



- **Retail chase recent moves,** and the price reverts after them.

Timing: Price Impact Declines Most Sharply at High Leverage



Every Cohort Loses, Each for a Different Reason

- **Execution cost is the same everywhere;** price impact falls with leverage.
- **Low leverage loses through liquidation,** mostly on October 10.

COHORT	EXECUTION COST (BPS)	PRICE IMPACT, 2 H (BPS)	LIQUIDATION	FEE (BPS)	TOTAL
Q1	−2.38	**+2.38**	**−$66.8M**	−2.87	−$90M
Q2	−2.51	+2.90	−$77.0M	−2.83	−$107M
Q3	−2.70	+2.05	−$44.3M	−2.93	−$112M
Q4	−2.17	+1.19	−$33.1M	−3.19	−$107M
Q5	−2.06	**+0.31**	−$6.6M	**−3.71**	−$117M

Two Levers Would Cut Most of the Loss

Retail lose, but protecting them is not that hard

- **Fees:** a cap on trading fees would cut the largest channel.
- **Leverage limits:** the 2010 U.S. retail FX cap cut high-leverage losses by 40%.

Thank you

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